L10: Cascades in Networks with Communities

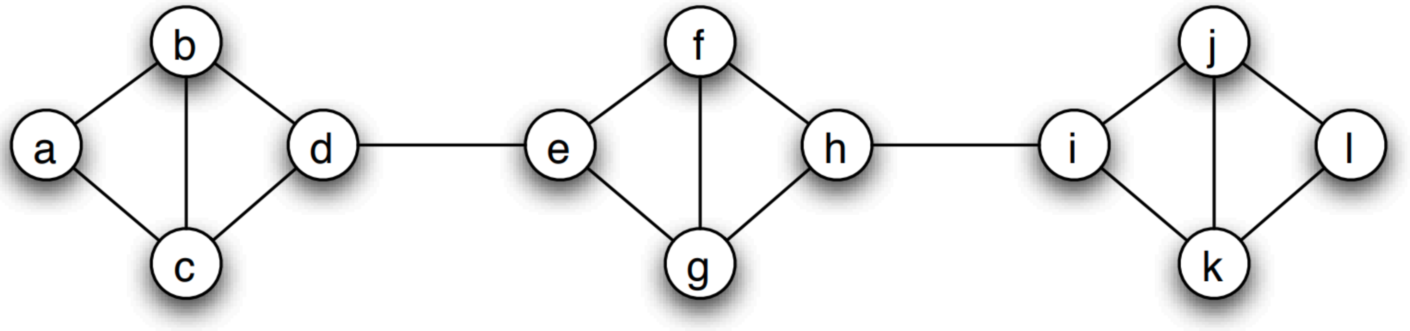

Image source: Textbook [***"Networks, Crowds and Markets"***](https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf) ***(https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf)*** *book, by Easley and Kleinberg, Chapter 19*

We have already established in earlier lessons that most real-world networks have clusters of strongly connected nodes that we refer to as *communities*.

How do such communities affect the extent of diffusion processes, such as those discussed earlier in this lesson?

Is it that the community structure facilitates or inhibits the diffusion of information on networks?

To make this analysis more concrete, let us start with a quantitative definition of a network community structure:

We say that a set of nodes *C* forms a cluster *(or community)* of density *p* if every node in that set has at least a fraction p of its neighbors in the set *C*.

The visualization shows three such clusters of nodes, each of them containing four nodes, with density *p=2/3*.